

NAMAN PARAS VORA

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EDUCATION

Clemson University (CUICAR), Greenville, SC

Aug 2025 – Aug 2027

M.S. Automotive Engineering | GPA: 3.91 / 4.00

Vellore Institute of Technology

B.Tech. Mechanical Engineering | GPA: 9.05/ 10.00

July 2019 - May 2023

ENGINEERING ACHIEVEMENTS

- Composite Design and Lightweighting:** Optimized a CFRP-foam skateboard to **364 g**, limiting deflection to **4 mm under a 1,200 N load**, through FEA and topography optimization. Created **GD&T-based SolidWorks engineering drawings** for prototyping (DFMPro) & VARTM fabrication. (CUICAR)
- Biomimetic Structural Design:** Engineered and validated interlocking bike frame design based on Ironclad Beetle exoskeleton, modeled load-transfer features that demonstrated **46.4% higher flexural load bearing capacity** than solid baseline specimens. (CUICAR)
- Battery Thermal Analysis:** Simulated performance of 12 nanofluids across **720 ANSYS Fluent cases**, identifying **Cu and Au for best heat-transfer performance** and **Al₂O₃ and Ag for thermal stability** for battery cooling in electric vehicles. (Simulation Lab)
- CFD and Aerodynamic Performance:** Evaluated surface changes in STAR-CCM+ on a closed wheel vehicle, successfully reducing drag coefficient from **0.51 to 0.34** by optimizing the greenhouse angle. (Deep Orange 18, CUICAR)
- Manufacturing & Dimensional Quality:** Oversaw export chassis dimensional accuracy and CMM analysis, helping achieve **<16 defects/1,000 vehicles** and **>92% straight-pass ratio** for 2 assembly lines. (Honda)
- Published Research (Rapid Prototyping):** Co-authored "Performance evaluation of 3D printing technologies: a review, recent advances, current challenges, & future directions," DOI: [10.1007/s40964-021-00257-4](https://doi.org/10.1007/s40964-021-00257-4) (VIT)

PROFESSIONAL EXPERIENCE

Clemson University - Deep Orange 18 | Body & Structures Engineer

Jan 2026 – Present

- Designed tube frame chassis concept with modular greenhouse in **Siemens NX** and **HyperMesh** for a 200+ mph autonomous vehicle based on Next Gen NASCAR architecture.
- Performed modal and NVH analysis to identify low-vibration mounting locations for autonomy sensors and support structural integration and sensor reliability; designed floor panel with mode shape frequencies > 40Hz by altering bead parameters through topography optimization.

FIA Formula 4 Championship, India | Mechanical Engineer Intern

Aug 2024 - Oct 2024

- Collaborated with race engineers and drivers on vehicle setup and trackside execution, including front suspension rebuilds, wing-angle adjustments, brake bias, and tire-pressure tuning.
- Analyzed onboard video and telemetry using RaceStudio and WinTAX, including braking, throttle, RPM, and lap traces, contributing to a **2-second lap-time improvement** and **2 podium finishes**.

Simulation Lab, India | CAE Engineer

Jul 2024 – Sept 2024

- Studied EV Battery Thermal Management Systems, evaluating nanofluid cooling performance by monitoring module inlet-outlet temperatures under varying battery surface temp (30 – 35°C) and thermoelectric-cooler temperatures (2 – 18°C).
- Built and analyzed ANSYS Fluent models to simulate coolant performance and aid in material selection.

Honda Motorcycle & Scooter India | Executive Trainee, Production Body

Jul 2023 - Jul 2024

- Supported dimensional validation and **production launch from PP1 phase through mass-production** in collaboration with New Model Development team, while overseeing export chassis production for BSIV and BSVI CKD/ CBU versions.
- Led a 23-member team in Press & Weld supporting production of 5,000+ vehicles/day, collaborating with Assembly Frame and Vehicle Quality teams to **control defect occurrence and outflow**.
- Collaborated with chassis suppliers and vendors to apply 7 QC tools for defect investigation and drive root-cause corrective actions and Kaizen.

PROJECTS

Battery Pack Modeling & Powertrain Sizing | Clemson University – CUICAR

- Sized series-parallel hybrid powertrains from measured drive-cycle demands for passenger vehicle applications.
- Modeled **battery SOC, OCV, internal resistance, heat generation, and coolant response** in Simulink, maintaining pack temperature < **32°C**
- Validated powertrain performance over drive cycles, achieving 90 kW combined hybrid output through optimized engine-motor operation and regenerative braking.

Multivariant Transmission Assembly Line Design Using Siemens Plant Simulation | Clemson University – CUICAR

- Designed and optimized a transmission line for 6,8 and 9 speed variants in Siemens PlantSim for a 70,000-unit annual target.
- Reduced machine waiting time by 30.04% and annual operating cost-saving of 53.3% through optimizing layout, material flow and shift strategy.

TECHNICAL SKILLS

CAD & Mechanical Design: Siemens NX, SolidWorks, Fusion 360, AutoCAD, GD&T, DFMPro, 2D Engineering Drawings

FEA & CFD: HyperMesh/OptiStruct, ANSYS Mechanical, Fluent, STAR-CCM+, LS-DYNA

Prototyping & Manufacturing: 3D printing, VARTM, CMM analysis, Siemens PlantSimulation, Lean Six Sigma Green Belt

Modeling and Data Analysis: MATLAB, Simulink, CarSim, MS Office, Minitab, WinTAX, TRNSYS